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ORIGINAL PAPER

Developing vehicular traffic noise prediction model through ensemble machine learning algorithms with GIS

Ahmed Abdulkareem Ahmed[1] · Biswajeet Pradhan[1,2] · Subrata Chakraborty[1] · Abdullah Alamri[3]

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Abstract

Vehicular traffic noise is an important aspect that requires environmental impact assessment (EIA) due to the increase in vehicular traffic. This has called for the need for detailed assessment, quantification and modelling of traffic noise pollution among industrial and scientific communities. This study uses an ensemble of machine learning algorithms to enhance the vehicular traffic noise prediction. The noise prediction is performed for the noise measure of the equivalent continuous noise level per 15 min ($L_{eq,\ 15\ min}$) along the Subang Jaya and Shah Alam New Klang Valley Expressway (NKVE). In this study, three machine learning methods—(i) artificial neural network model (ANN), (ii) correlation-based feature selection with artificial neural network model (CFS-ANN) and (iii) ensemble machine learning algorithms random forest with artificial neural network model (Ensemble RF-ANN) are deployed to estimate the $L_{eq,\ 15\ min}$ during the peak hours of the morning (6:30–8:30 a.m.), evening (6:00–8:00 p.m.) and night (10:00 p.m.–12:00 midnight) for the weekday (Monday). The root mean square error (*RMSE*) and coefficient of determination (R^2) were used to select the best model. The prediction maps for morning, evening and night are prepared using geospatial modelling for the two sites under consideration. The results showed that the Ensemble RF-ANN model is the best model recording the lowest *RMSE* during the training and testing with values of 1.767 and 2.378, respectively. It also achieved the highest R^2 values of 0.923 and 0.835, respectively, for the training and testing. The study provides novel noise models and excellent analysis of results capable of identifying key factors affecting noise in a geographical area. The study outcomes could be utilised by noise modelling consultants, town planners and traffic modelling experts for traffic noise mapping of larger geographical areas.

Keywords Vehicular traffic noise · GIS · Remote sensing · Ensemble machine learning · Correlation-based feature selection · Artificial neural network

Introduction

Vehicular traffic noise is a very important area of interest in environmental impact assessment (EIA) (Gilpin 1995; Ortolano and Shepherd 1995; Lohani et al. 1997; Thomas 1998; Glasson and Therivel 2013; Ruiz-Padillo et al. 2016; Ahmed and Pradhan 2019; Alkheder and Almutairi 2021; Singh et al. 2021). Because of the rapid urban expansion especially road networks, it has invariab$_l$y become a necessity to study vehicular traffic noise in most developing countries, particularly in Malaysia. Public health and urban planners have made it mandatory to provide a detailed assessment, quantification and model of traffic noise pollution around communities due to its effect on residents (Babisch 2002; Murphy and King 2014; Lee et al. 2014; Roswall et al. 2015; Khaiwal et al. 2016; Fisher et al. 2018; Paiva et al. 2019; Ky et al. 2021; Rahman et al. 2021).

Over the past decade, research works on machine learning (ML) domain have focused more on the new methods of constructing ensembles, with the artificial neural network (ANN) being one of the major approaches used in them. The ANN

Responsible Editor: Amjad Kallel

✉ Biswajeet Pradhan
Biswajeet.Pradhan@uts.edu.au; biswajeet24@gmail.com

[1] Centre for Advanced Modelling and Geospatial Information Systems (CAMGIS), Faculty of Engineering and Information Technology, University of Technology Sydney, Sydney, Australia

[2] Earth Observation Centre, Institute of Climate Change, Universiti Kebangsaan Malaysia, 43600 Bangi, Malaysia

[3] Department of Geology and Geophysics, College of Science, King Saud University, Riyadh 11451, Saudi Arabia

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method has demonstrated better performance and accuracy when compared with multiple linear regression methods (Tan and Gilbert 2003; Erdem et al. 2005; Polikar 2012; Zhou 2012; Heinermann and Kramer 2016; Krawczyk et al. 2017; Fox et al. 2017; Pham et al. 2017; Lieske et al. 2018; Medlock and Papin 2020; Kim et al. 2021).

In 2010, Givargis and Karimi (2010) carried out hourly A-weighted equivalent sound pressure levels ($LA_{eq,1h}$) applying the ANN for five roads in Tehran. The results indicated that traffic noise can be predicted accurately in a statistical manner using a neural network approach. No substantial variance was observed between the absolute testing set errors of the neural network and the Calculation of Road Traffic Noise (CoRTN) model (Begou and Kassomenos 2021) (calibrated version). Similarly, Kumar et al. (2014) predicted 10 percentiles sound level (L_{10}) and the equivalent continuous sound level (L_{eq}) in decibels (A) using the ANN approach for vehicular traffic noise assessment in India. The parameters considered in the ANN model comprise the hourly vehicle volume, average vehicular speed, and proportion of heavy vehicles. The outcome was also compared with a multiple linear regression model with the ANN approach, achieving more reliable results for traffic noise modelling.

In 2015, Garg et al. (2015) investigated traffic noise prediction in Delhi, India, and compared the performance between the outcomes of the ANN model and multiple linear regression model. The ANN model produced more accurate and reliable results than the multiple linear regression model in their study. However, both the root mean square error (*RMSE*) and the coefficient of determination (R^2) of the measured and predicted data were observed to be optimised by ANN. A recent road traffic noise model was carried out by Hamad et al. (2017) in the hot climate of Sharjah City in United Arab Emirates using ANN. The ANN model was designed to include the traffic classification, volume, average speed, and pavement temperature along three different routes in Sharjah City as input parameters. It was observed that the ANN model outperformed the other two conventional models (the Ontario ministry of transport road traffic noise model, ORNAMENT, and the basic statistical traffic noise (BSTN) model). Furthermore, the pavement temperature parameter was observed to improve both the ANN and conventional road noise models.

In this study, we compare the performance of the artificial neural network model (ANN), the correlation-based feature selection with artificial neural network model (CFS-ANN) and the ensemble machine learning algorithms random forest with artificial neural network model (Ensemble RF-ANN) to measure the equivalent continuous noise level per 15 min ($L_{eq,15\ min}$) during peak hours of the morning (6:30 a.m.–8:30 a.m.), evening (6:00 p.m.–8:00 p.m.) and night (10:00 p.m.–12:00 midnight) of the weekday (Monday). The traffic data was acquired along the NKVE expressway using field survey and other methods such as Light Detection and Ranging (LiDAR) and Worldview-3 image. The models were validated using both the *RMSE* and R^2 measures. The novelty of this study lies in the development of the ensemble machine learning algorithms by combining the random forest model with the ANN model (Ensemble RF-ANN). Also, the output variables of the proposed model (Ensemble RF-ANN model) and the other compared models (ANN and CFS-ANN models) are displayed as three mapped layers: traffic noise in the morning, evening and night. Furthermore, models utilised more variables including humidity, temperature, road density, digital surface models (DSM) and gradient of the location, time of the day and more specific vehicle classes (light vehicle, heavy vehicle, motorbike, truck and lorry, bus).

The key research questions we aim to resolve in this study include the following: (i) Can the RF-ANN ensemble outperform popular standard ANN and variant CFS-ANN models? (ii) How does incorporating additional variables impact the noise modelling? and (iii) How can noise prediction maps be generated using such models?

Study location and data collection approach

Study location

The New Klang Valley is a densely populated area in Malaysia which is located along the NKVE expressway as shown in Fig. 1; it was chosen as the study location. It is a 35-km-long expressway running between Jalan Duta in Kuala Lumpur and New Klang urban/industrial area (Bukit Raja). It is a very busy route for the residents of Kuala Lumpur, Shah Alam, Petaling Jaya, Damansara, Sungai Buloh, Subang Jaya and Klang. The stipulated speed limit for the expressway is between 90 and 110 km/h. In this study, two sites, Subang Jaya and Shah Alam, were carefully selected along the NKVE expressway to achieve the objectives of the investigation. The selected areas exhibit numerous suitable features, such as residential, commercial and industrial areas, as shown in Fig. 1. These sites are suitable for traffic noise investigation.

Data collection

Continuous measurement of noise levels at 15-min intervals (minimum, maximum and average) was carried out using a type-A filter (dB (A)) with the aid of Sound Level Meter TES-52 (0.1 dB resolution). The measurement devices were mounted at locations 30 cm from signs and 2 m away from noise barriers. Garmin Global Positioning System (GPS) device GPS 60 determined the coordinate of each location. Three measurements were taken daily during the weekday (Monday) in February 2017: morning (6:30 a.m.–8:30 a.m.), evening (6:00 p.m.–8:00 p.m.) and night (10 p.m.–12

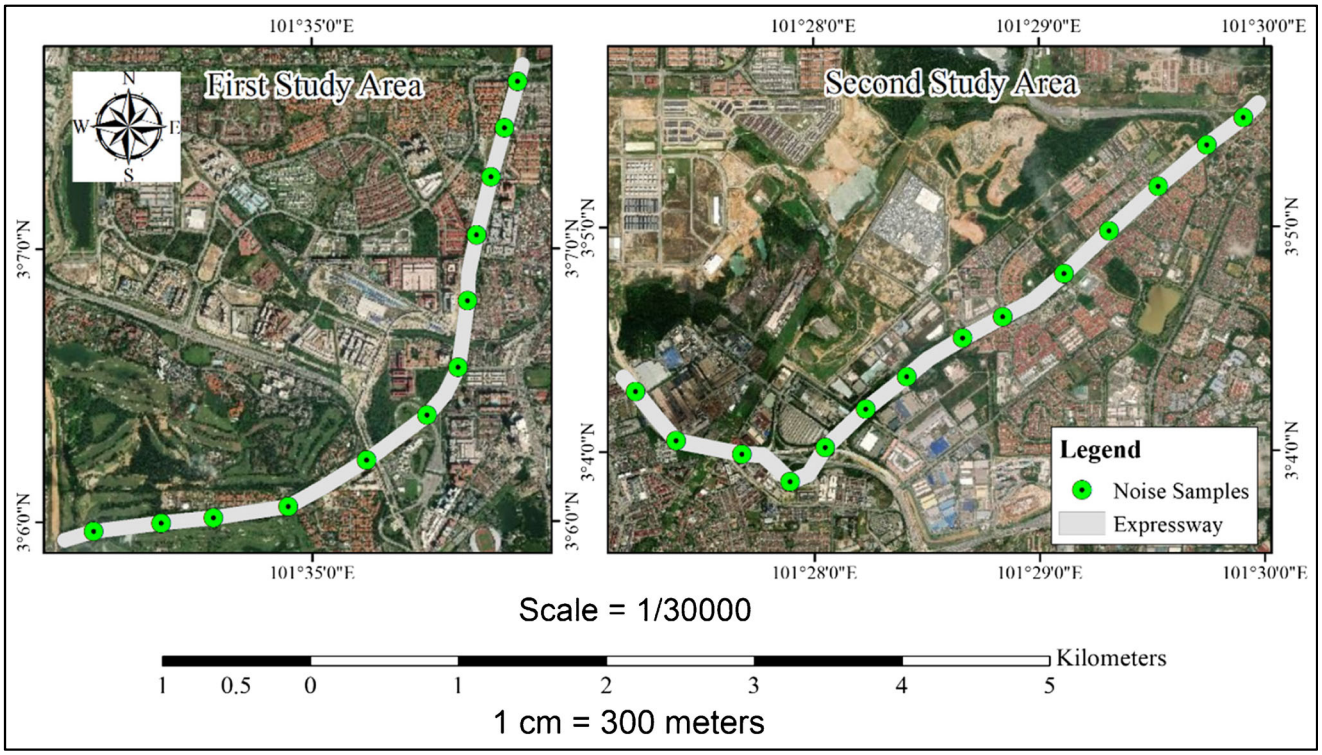

Fig. 1 The two sites of the study area: Subang Jaya and Shah Alam

midnight). The traffic flow information was collected in the field using manual recording and checked by a video recorder. Finally, GIS was used for the road information acquisition such as DSM, gradient, density of road and the location of noise samples. These parameters are important for vehicular traffic noise prediction, modelling and mapping. The parameters (noise predictors) used in this study were noise $L_{eq,15\ min}$, light vehicle, heavy vehicle, motorbike, truck and lorry, bus, DSM, time, gradient, density of road, temperature, humidity and location of noise samples as shown in Table 1.

Methodology framework

The methodological framework for the study is shown in Fig. 2. First, data collection is done using the field survey for subsequent management in GIS software. The noise samples were collected at two different locations in the study area. The study area sites (Subang Jaya and Shah Alam) are carefully selected from the NKVE of the study area. Equivalent continuous noise level per 15 min ($L_{eq,\ 15\ min}$) at the study area was measured. The data set (78) was separated into two sets, 70%

Table 1 Summary statistics of noise predictors

Noise predictors	Minimum	Maximum	Mean	Deviation
Noise $L_{eq,\ 15\ min}$	71.55	103.44	90.6607	6.07889
Light vehicle	7	2820	833.415	804.569
Heavy vehicle	2	618	81.6748	117.387
Motorbike	0	605	92.0407	109.277
Truck and lorry	2	443	55.7805	82.026
Bus	0	175	25.8943	35.9464
DSM	3.7425	41.61	17.4088	11.1514
Time	Morning, evening and night			
Gradient	0.395279	203.846	12.2455	32.5366
Density of road	0.034901	0.051338	0.0474467	0.00225269
Temperature	19.8	28.8	23.7837	2.88857
Humidity	42.7	73.4	61.022	7.7833
Location of noise samples	*XY* coordinate system			

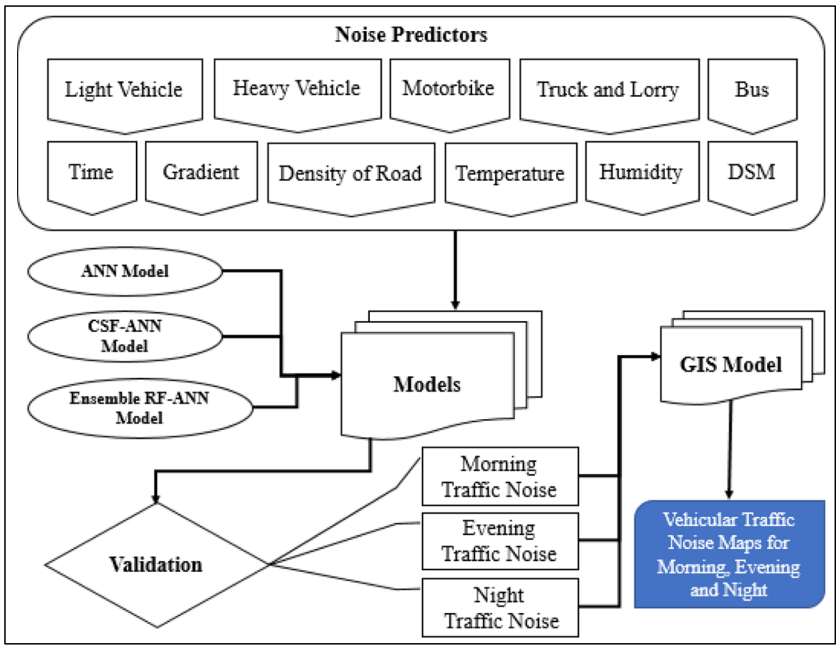

Fig. 2 The methodology framework

of training data (54) and 30% of testing of data (24). Three models, ANN, CFS-ANN and Ensemble RF-ANN, were developed and applied with the dataset. *RMSE* and R^2 measures were used to select the best model. Finally, prediction maps for morning, evening and night were prepared for these two sites of the study area through the use of geospatial modelling based on GIS.

The ANN model

ANN is one of the most commonly used machine learning models that uses statistical- and biological-inspired learning models. It is deployed in models to develop systems state using nonlinear combinations of various input variables (Recknagel 2001; Cruz and Wishart 2006; Goyal et al. 2014; Patel et al. 2015; Witten et al. 2016; Wang and Srinivasan 2017; Charte et al. 2018; Goldstein et al. 2019). Back-propagation network was used in our study with sigmoid activation functions incorporated in the hidden layers. Also, this architecture used linear activation function applied in the output nodes with only one hidden layer. This approach is based on the report by Bishop (1995) who inferred that ANN with more than one hidden layer is not necessary. This ANN architecture is trained with a back-propagation algorithm with momentum and gradient descent.

In ANN, it is required that the number of nodes found in a single hidden layer, learning rate and the maximum number of training epochs are identified (Hill and Minsker 2010). Therefore, the optimal number error was used in this study coupled with variation of the hidden layer number between 3 and 30 and learning rate between 0.01 and 0.8 at an incremental level of 0.05. Table 2 shows the hyperparameters used in the proposed ANN model for the prediction of traffic noise in the search space. R^2 and *RMSE* were measured for each configuration. Subsequently, the model data were measured, and the output calculated. Generally, the ANN algorithm is designed based on the error minimisation principle through successive gradient as shown in Eq. (1). The ANN model has brought great potential for remote sensing–related applications. However, it is computationally intensive and has a tendency to overlearn (Baczyński and Parol 2004).

$$E = \frac{1}{2}\sum_{i=1}^{L}\left(d_j - o_j^M\right)^2 \tag{1}$$

where o_j^M and d_j denote the expected current response and output from the output layer node "j", respectively. "L" refers to the number of nodes utilised in the output layer. However, corrections of parameter weights are made and added to the previous weight values in an iterative manner as shown in Eq. (2):

$$\begin{cases} \Delta w_{i,j} = -\mu \dfrac{\partial E}{\partial w_{i,j}} \\ \Delta w_{i,j}(t+1) = \Delta w_{i,j} + \alpha \Delta w_{i,j}(t) \end{cases} \tag{2}$$

where the weight parameter between nodes i and j is represented by $w_{i,\,j}$. Δ represents the positive constant which regulates the adjustment level referred to as learning rate. α is denoted momentum factor with values between 0 and 1 while "t" refers to the number of iterations. Parameter α is called smoothing or stabilising factor (Yang 1995).

Table 2 Hyperparameters of the model for the traffic noise prediction and the search space used for the fine-tuning

Hyperparameters	Domain
Network class	{MLP}
No. of hidden units	(3–30)
Training algorithm	{BFGS, RBFT}
Hidden and output activation	{Identity, Logistic, Tanh, Exponential, Gaussian}
Rate of learning	(0.01–0.8) by step of 0.05
Momentum	(0.1–0.9) by step of 0.1

The correlation-based feature selection (CFS) with ANN model (CFS-ANN)

The process of subset selection of features/predictor variables that influences the performance of a model in terms of accuracy is termed feature selection. Usually, the criteria used for the selection involves minimisation of performance measure of a specific model such as mean squared error. Literarily, the data dimensionality of the feature selection is reduced through elimination of redundancy to best fit the model. The CFS algorithm with ANN would be used for this study.

The CFS model is well known as a filter algorithm applied in feature selection processes that rely on the correlation function. This algorithm involves the selection of subcategories that comprise features that are highly correlated to a specified class but not necessarily related to each other. Conversely, features with low correlation class are usually exempted. Also, the features that are repetitive are checked out due to the possible relationship that might exist with another feature. The acceptability of a specific feature is based on its ability to predict classes within an instance space not as expected by distinct features. The CFS feature subset is based on Eq. (3):

$$Ms = \frac{k\overline{r_{cf}}}{\sqrt{k + k(k-1)\overline{r_{ff}}}} \tag{3}$$

where Ms represents the heuristic "merit" of a given feature subset s consisting of k distinct features, $\overline{r_{cf}}$ represents the mean value of the feature-class correlation (f 2 s) and $\overline{r_{ff}}$ denotes the average of the feature to feature inter-correlation.

The ensemble machine learning algorithm (Ensemble RF-ANN model)

Ensemble algorithms are a powerful tool that combines predictions from multiple machine learning models (Zhou 2012). Ensemble learning aids in terms of machine learning output improvement through the combination of several models and has been applied in noise classification (Xiao 2019). This method enables better prediction performance when compared with a single model. In this study, ensemble machine learning algorithms integrated with random forest and ANN were used.

Breiman (2001) proved that the random forest (RF) algorithm is a powerful approach and relatively newly used for regression, classification and unsupervised learning (Liaw and Wiener 2002). Since then, many fields have seen applications of this approach to improve performance (Gupta et al. 2015; Youssef et al. 2015). When deploying regression or RF in classification, a variable set known as training dataset is selected randomly from the entire dataset using the bagging method to ensure model calibration (Chen et al. 2014). When variable values are permuted across the out-of-bag observations, the function obtains the model prediction error for each variable (Trigila et al. 2015).

Another machine learning algorithm that trails the bagging technique is the random forest. It is surely an extension of the bagging estimator algorithm. The decision trees are the major estimators incorporated in the RF. It is dissimilar to the bagging meta estimator, because random forest selects sets of features randomly to be used for the decision of the best split at every nodal point of the decision tree. The steps in RF algorithm are as follows: RF subsets are produced from the initial dataset (bootstrapping) at every nodal point in the developed decision tree. So, only the random feature set is then considered in deciding the best possible split. The decision tree is built-in for each of the decision subsets before final predicted values are estimated by taking the average of the predictions from all the decision trees.

The GIS model

The GIS model was spatially designed to represent the traffic noise prediction level generated from vehicular traffic on the expressway. The design is based on the final implementation of the proposed model (best model). The parameters used in the model were converted to a geodatabase through sample connection of the attributes with their various positions obtained with the aid of the Global Positioning System (GPS). Further conversion of the parameters was done to raster form using geostatistical interpolation (inverse distance weighted (IDW) interpolation) for proper information on noise prediction. The IDW method was chosen due to its ability to provide a highly correlated result than the spline and kriging methods.

On the other hand, higher distortion of the interpolated result was observed in the spline and kriging results compared with the IDW. The model's parameters were combined in GIS background based on the overlying inquiries of the proposed model. It was applied on a grid at a high resolution of 5*5 m in order to predict road traffic noise for the unsampled areas. Each grid exhibits values based on the intersected values that represent the disparity in traffic noise values and distribution of traffic noise level in the results.

Model evaluation

The predictive performances were measured by determining the potentiality and effectiveness of the three models through an estimation of the R^2 and *RMSE* values for $L_{\mathrm{eq},\ 15\ \mathrm{min}}$. The performance measures above indicate the level of accuracy of each prediction through comparison of the actual value of parameters (a_i), the predicted values (b_i), the data points of the sample (n) and other variables like average observed values ($\overline{a}$) and average predicted values ($\overline{b}$) which can help during comparison with different models.

Equation (4) is used to assess the correlation (r) for the linear relationship between the two data sets. Equation (5) estimates the coefficient of determination R^2 which is the proportion of variance of the dependent variable to the total variance by the regression model. Usually, R^2 are values in the range of 0 to 1. This represents fitness of data on the regression line. An R^2 value of 1 indicates a perfect model, while a zero (0) shows no variance. Furthermore, Eq. (6) is used to estimate the *RMSE* for evaluation of the average performance across different testing samples.

$$r = \frac{\sum_{i=1}^{n}\left(a_i - \overline{a}\right)\left(b_i - \overline{b}\right)}{\sqrt{\sum_{i=1}^{n}\left(a_i - \overline{a}\right)^2 \sum_{i=1}^{n}\left(b_i - \overline{b}\right)^2}} \tag{4}$$

$$R^2 = r^2 \tag{5}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}\left(|bi - ai\ |\right)^2}{n}} \tag{6}$$

Results and discussion

Evaluating the contribution of the noise predictors using the CFS algorithm

In this study, the noise predictors' contribution by the CFS algorithm (CFS-ANN model) is presented. The proposed model parameters exhibit different levels of contribution for prediction of road vehicular traffic noise in the two sites of the study area. In the CFS analysis, eight parameters were selected instead of 12, which is based on the effectiveness of predictors. These predictors which were then used in the CFS-ANN model were as follows: light vehicle, heavy vehicle, motorbike, DSM, time, gradient, humidity and location of noise samples. Based on the statistical analysis, these parameters exhibited the best performance for traffic noise prediction. Generally, the predicted traffic noise highly correlates with the light vehicle, heavy vehicle, time and location of noise samples.

Results of ANN, CFS-ANN and Ensemble RF-ANN models

Figures 3, 4 and 5 and Table 3 show that Ensemble RF-ANN is the best model, because it has the least *RMSE* value of training (1.767) and testing (2.378), respectively. However, this proposed model exhibited the highest correlation coefficient (R^2) of 0.923 and 0.835 for training and testing, respectively, whereas the *RMSE* of training for ANN and CFS-ANN models are 3.713 and 2.659 and the R^2 of training for ANN and CFS-ANN models are 0.86 and 0.881, respectively. Also, the *RMSE* and the R^2 of testing ANN and CFS-ANN models are (5.589, 3.272) and (0.707, 0.688), respectively.

Sensitivity analysis of the Ensemble RF-ANN model

In this section, the influence of the different parameters on the performance of the ensemble RF-ANN model is discussed. The parameters have a significant influence on the accuracy of the models for predicting traffic noise. The parameters we tested include the learning rate, hidden layer units, gradient momentum value and the activation function applied.

Firstly, the number of hidden units in the network has a significant effect on the prediction capacity of a model. Sensitivity analysis carried out on these parameters indicated 21 hidden units as the best value for use in this model, as shown in Fig. 6. The R^2 and *RMSE* achieved with the 21 hidden units are 0.923 and 1.767, respectively. However, larger or smaller number of units could decrease or increase the *RMSE* and R^2 values of the model. For instance, the use of 9 hidden units increased the *RMSE* value to 7.124, while R^2 reduced to 0.795. Also, the use of 30 units achieved an *RMSE* value of 6.261 and R^2 of 0.773.

Secondly, the influence of the gradient momentum and the learning rate on the performance and optimisation of the model is studied as shown in Figs. 7 and 8. The best value achieved with learning rate of 0.55, with an *RMSE* and R^2 of 1.767 and 0.923, respectively. A small learning rate value (0.01) decreased R^2 to 0.824 and increased *RMSE* to 5.218. Also, when high values (0.7 and 0.9) were used, *RMSE* were 5.697 and 4.168, respectively, while R^2 values were 0.834 and 0.786.

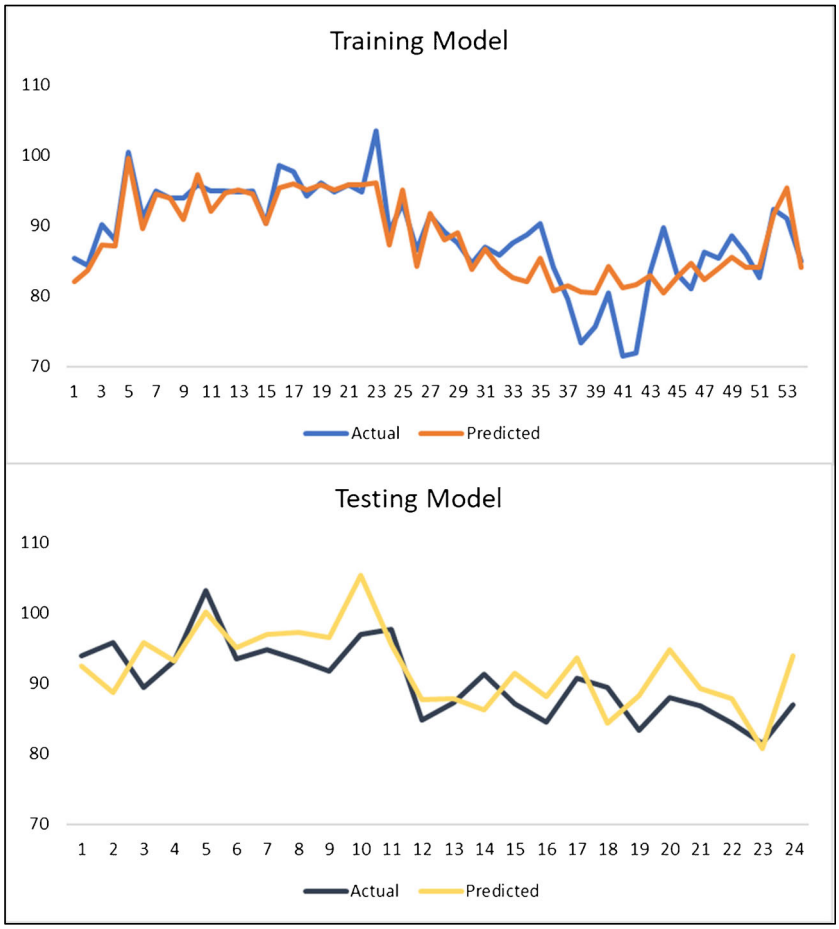

Fig. 3 R^2 of training and testing for the ANN model

Furthermore, considerable time is required for the algorithm to converge. However, when the learning rates are large, the algorithm could step over a minimum, causing oscillation which results in unstable learning. On the other hand, increasing the momentum value of the optimisation algorithm in turn improves the *RMSE* with R^2 of noise prediction model. However, the small values of momentum decrease the value of R^2 and increase the value of *RMSE* in the proposed model, whereas the *RMSE* with R^2 of the model was enhanced from 0.5 (*RMSE* = 2.462 and R^2 = 0.861) to 0.9 (*RMSE* = 1.767 and R^2 = 0.923). This shows that gradient momentum is highly required to avoid being stuck at the local minima. Generally, a larger momentum value enables faster convergence. But, small momentum values cannot reliably remove the local minima phenomena and in turn may slow down the training of the model.

Finally, we analyse the impact of applying different activation functions and observe that this can significantly affect the performance of the RF-ANN model. We found that the identity activation function provides the best performance. The *RMSE* and R^2 of the proposed model achieved 1.767 and 0.923, respectively. By contrast, the exponential function recorded the worst for the hidden and output layers, whereas the *RMSE* and R^2 recorded were 5.219 and 0.824, respectively, as shown in Fig. 9.

The geospatial model based on GIS based on the RF-ANN model

Noise maps were generated for the selected study locations using the proposed methods at different periods of time, including morning, evening and night at a weekday (Monday). The detailed maps are presented in Figs. 10, 11, and 12. The figures show the noise level prediction of morning, afternoon and night for the selected study areas (Subang Jaya and Shah Alam) on Monday. In Subang Jaya, the minimum noise level was 84.28, 93.57 and 80.03 dB for morning, evening and night, respectively. In addition, the maximum noise level of the morning, evening and night was 96.94, 100.36 and 103.41 dB, respectively. On the other hand, the noise level prediction shows that for Shah Alam, the minimum noise level recorded was 79.57, 71.62 and 72.04 dB in the morning, evening and night, while the maximum noise level was found to be 86.05, 92.63 and 94.18 dB for morning, evening and night,

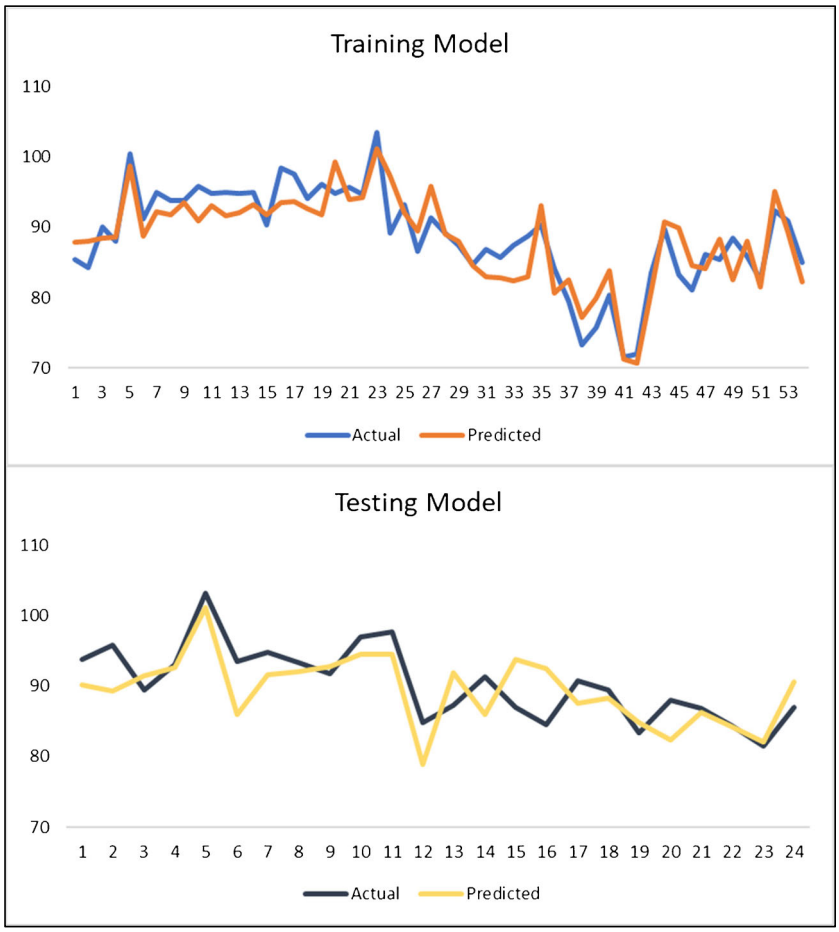

Fig. 4 R^2 of training and testing for the CFS-ANN model

receptively. Figure 10 shows traffic noise in the morning, Fig. 11 for the evening and Fig. 12 for night. Based on these figures, the traffic noise level in the first study area (Subang Jaya) is higher than the traffic noise level in the second area (Shah Alam).

Conclusion

We developed an ensemble model that maps vehicular traffic noise (equivalent continuous noise level per 15 min ($L_{eq,\ 15\ min}$) for morning, evening and night at two designated areas along the NKVE expressway. This is because vehicular traffic noise is one of the main issues affecting the environment that needs to be addressed especially in densely populated areas where highways are present. Factors such as light vehicle, heavy vehicle, motorbike, truck and lorry, bus, DSM, time, gradient, density of road, temperature, humidity and location of noise samples are considered in the model. Models like ANN, CFS-ANN and Ensemble RF-ANN were used with noise predictors as input and equivalent continuous noise level per 15 min as output through nonlinear functions.

Though these models seem promising for the prediction of traffic noise, careful hyperparameter and optimisation are required to derive their utmost benefit. This study reveals the adequate fine-tuning of the Ensemble RF-ANN model can predict traffic noise levels accurately through error minimisation. The result achieved are *RMSE* (1.767 dB (A)) and the maximum of correlation coefficient (R^2) of 0.923. On the other hand, the vehicular traffic noise prediction highly correlated with the noise predictors such as the light vehicle, heavy vehicle, time and location of noise samples. These parameters gave the most significant traffic noise prediction for the CFS- ANN model.

The RF-ANN model was further analysed using sensitivity analysis of the ensemble parameters. The result shows that a larger quantity of units in the hidden layer may not necessarily lead to better accuracy of predictions than a lower quantity of units. Furthermore, the learning rate and the selection of appropriate activation function are necessary to achieve the desired accuracy level of noise predictions in the proposed

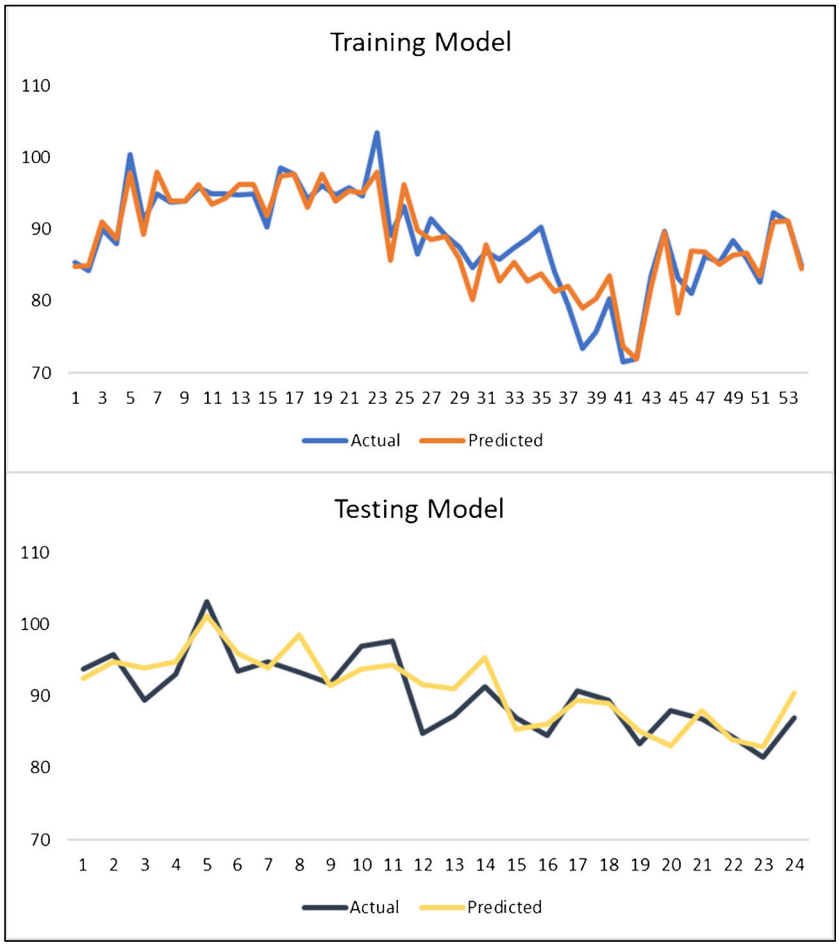

Fig. 5 R^2 of training and testing for the Ensemble RF-ANN model

model. The models were validated based on the *RMSE* and R^2 measures by comparing their calculated noise levels with the actual monitored noise levels. Finally, the noise prediction maps were prepared in GIS by using spatial, statistical and data management tools. The result shows that the model can be employed for noise distribution mapping on highways. This approach may offer more economical and time saving supportive mechanism to support traffic noise consultants. Future studies could be expanded in the areas of explainable AI models.

Table 3 Three best models determined for modelling vehicular traffic noise prediction

	ANN	CFS-ANN	Ensemble RF-ANN
Architecture model	12–26–3	6–11–3	12–21–3
RMSE of training model (error-dB (A))	3.713	2.659	1.767
RMSE of testing model (error-dB (A))	5.589	3.272	2.378
R^2 of training model	0.86	0.881	0.923
R^2 of testing model	0.707	0.688	0.835
Training algorithm	RBFT		
Hidden and output activation	Identity	Tanh	Identity

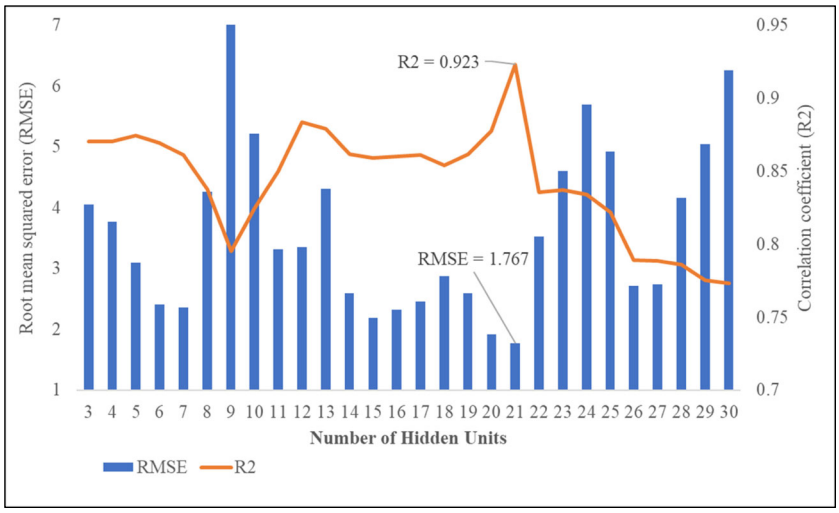

Fig. 6 Effects on the performance of the RF-ANN model based on the number of hidden units

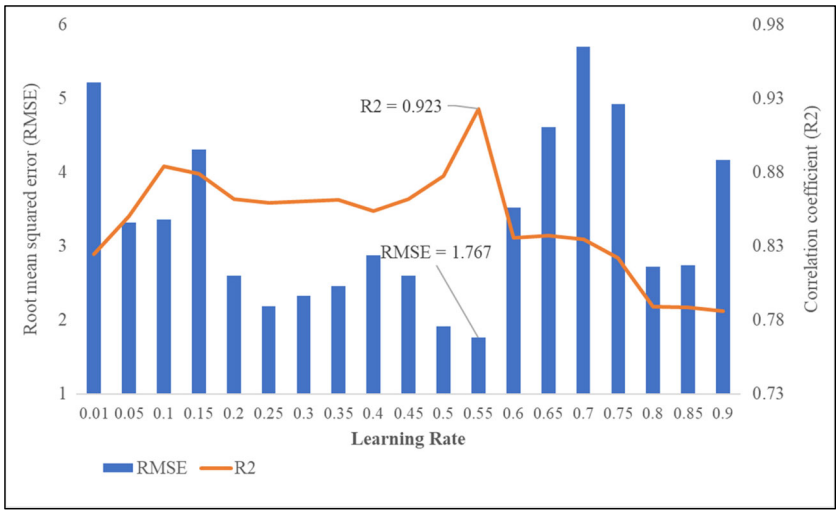

Fig. 7 Effects of the learning rate on the performance of the RF-ANN model

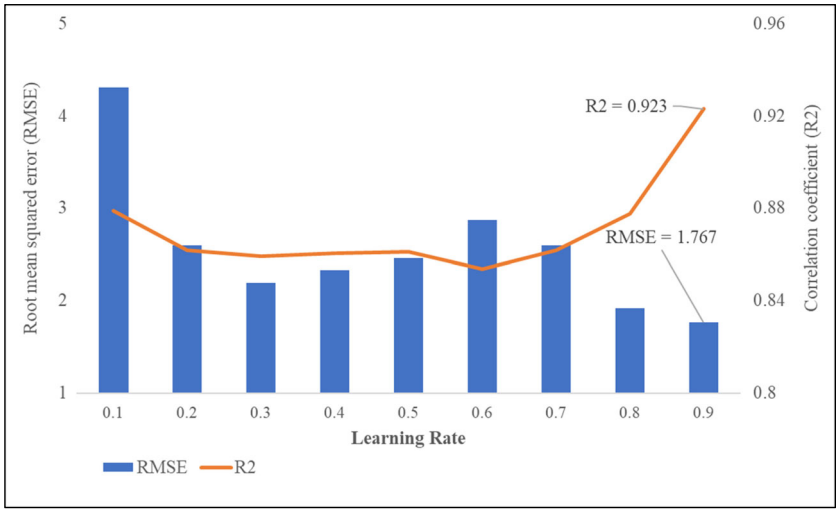

Fig. 8 Effects of the gradient momentum values on the performance of the RF-ANN model

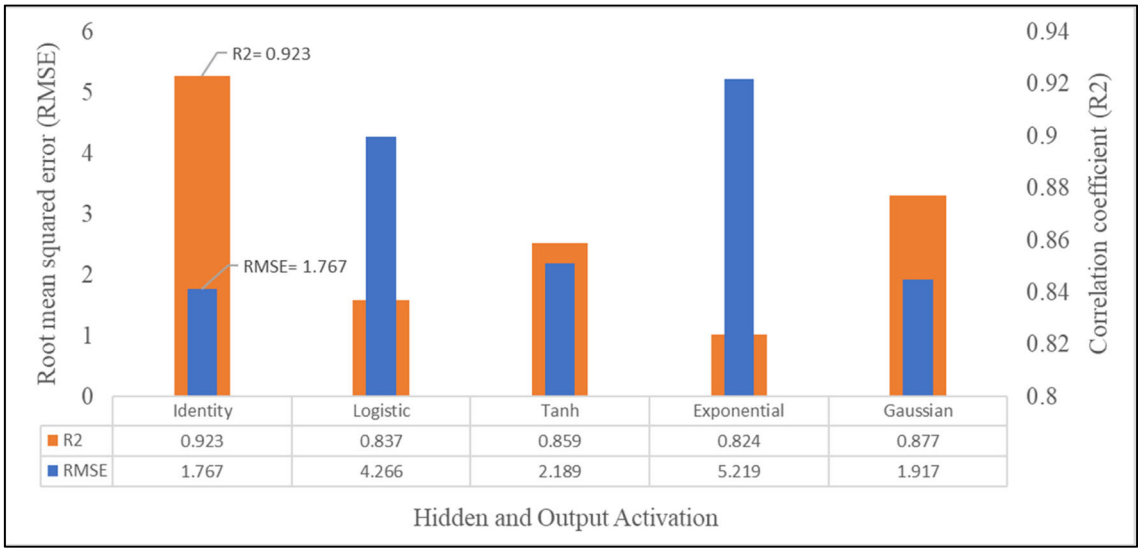

Fig. 9 Impacts of the various activation functions on the performance of the RF-ANN model

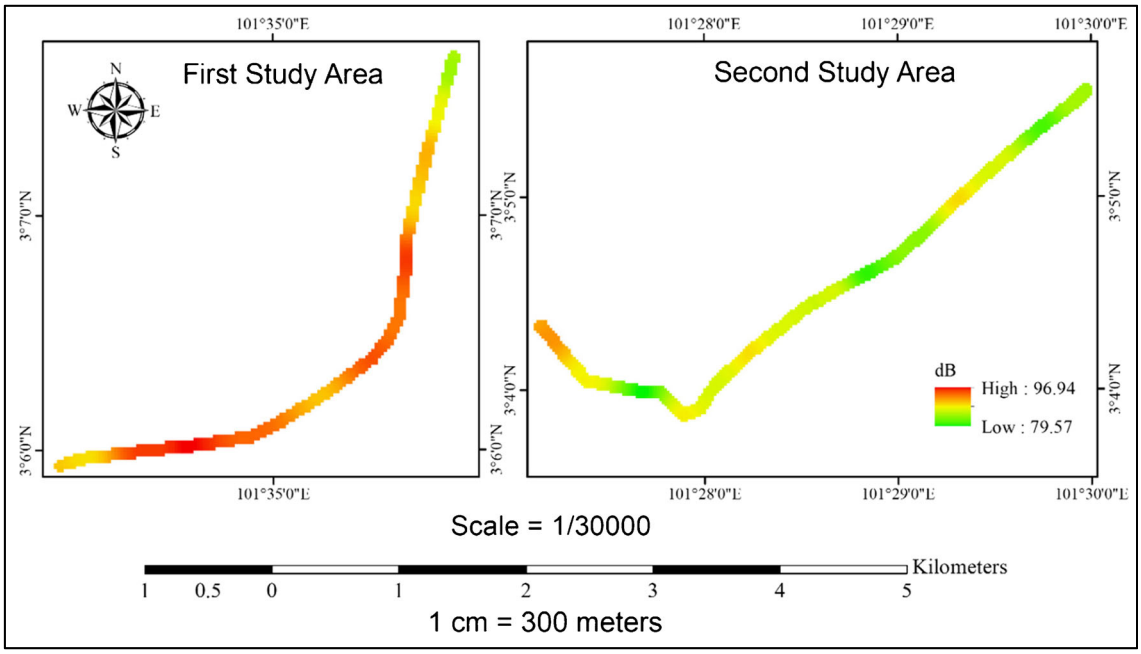

Fig. 10 Vehicular traffic noise level ($L_{eq,\ 15\ min}$) at the morning simulation in the study areas using the proposed model, whereas the first study area is Subang Jaya and the second area is Shah Alam

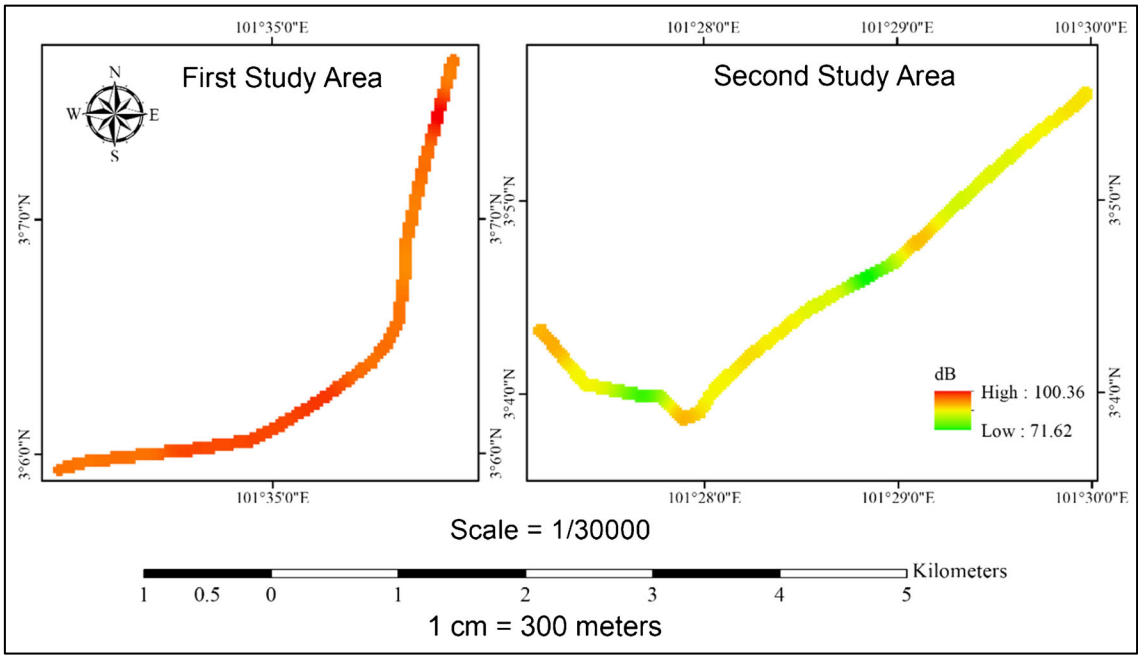

Fig. 11 Vehicular traffic noise level ($L_{eq,\ 15\ min}$) at the evening simulation in the study areas using the proposed model, whereas the first study area is Subang Jaya and the second area is Shah Alam

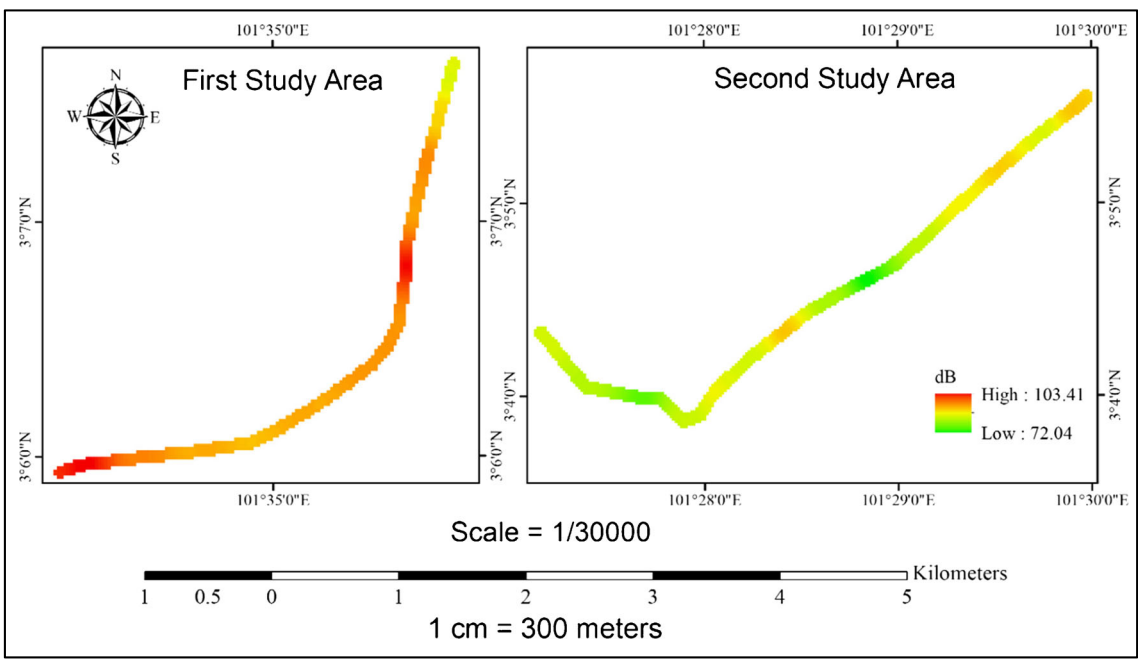

Fig. 12 Vehicular traffic noise level ($L_{eq,\ 15\ min}$) at the night simulation in the study areas using the proposed model, whereas the first study area is Subang Jaya and the second area is Shah Alam

Author contribution Conceptualisation, B.P.; overall supervision and concept, B.P.; methodology and formal analysis, A.A.; data curation, A.A.; resources, B.P.; writing—original draft preparation, A.A.; writing—review and editing, B.P., S.C., A.A.; supervision, B.P.; funding, B.P., A.A. All of the authors have read and agreed to the published version of the manuscript.

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Data availability The data can be made available from the corresponding author.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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